

Employee Management & Payroll System

Naming Convention Strategy

1. Tables : Table names will be written in lower case and in case of table name has two or more words then in camelCase without underscore and space.

Ex. Table :- employee or payrollDetails.

2. Column names : Column names will be written in lower case. Underscore (_) can be used if required.

a. Primary keys : primary keys will be named with table name + underscore (_) + id (keyword)

Ex. employee_id

b. Foreign keys : same as of primary keys but with respect to the corresponding table.

c. Stored procedures : Named with prefix keyword

sp + underscore + procedure name

Ex. sp_calcpayroll

d. Views : Named with prefix

v + underscore + view method name.

ex. v_attendancereport

Database Schema

Tables :-

1. Employee
2. Department
3. Designation
4. Attendance
5. Leave
6. SalaryStructure
7. Payroll
8. PayrollDetails
9. Role

Employee Management & Payroll System

1.Employee

Column	Type	Constraint
employee_id	Int	Pk
name	Varchar (50)	Not null
role_id	Int	Fk , not null
department_id	Int	Fk , not null
designation_id	Int	Fk , not null
manager_id	Int	Fk , can be null
email	Varchar (100)	Unique , not null
phone	Varchar (15)	Null
dob	Date	Null
password	Varchar(255)	Not null
Status	Varchar(10)	Not null, CHECK (Status IN ('Active','Inactive'))
gender	Varchar(5)	null

2.Department

Column	Type	Constraint
department_id	Int	Pk, not null
departmentName	Varchar(100)	Not null , unique
description	Varchar(200)	null

3.Designation

Column	Type	Constraint
designation_id	Int	Pk, not null
designationName	Varchar(100)	Not null , unique
description	Varchar(200)	null

4.Role

Column	Type	Constraint
role_id	Int	Pk , not null
roleName	Varchar(100)	Unique, not null

Employee Management & Payroll System

5.Attendance

Column	Type	Constraint
attendance_id	Int	Pk, not null
employee_id	Int	Fk
date	Date	Not null
status	Varchar(10)	Not null, CHECK (status IN ('Present','Absent','HalfDay','Leave'))

6.Salary Structure

Column	Type	Constraint
salaryStructure_id	Int	Pk, not null
employee_id	Int	Fk
basicSalary	Decimal(12,2)	Not null
hra	Decimal(12,2)	Default 0
allowances	Decimal(12,2)	Default 0
deductions	Decimal(12,2)	Default 0
effectiveFrom	Date	Not null
effectiveTo	Date	Null

7.Leave

Column	Type	Constraint
leave_id	Int	Pk, not null
employee_id	Int	Fk
startDate	Date	Not null
status	Varchar(10)	Not null , CHECK (status IN ('Pending','Approved','Rejected'))
endDate	Date	Not null
leaveType	Varchar(20)	Not null
reason	Varchar(200)	Null
approvedBy	Int	Null , Fk (employee_id)

Employee Management & Payroll System

8. Payroll

Column	Type	Constraint
payroll_id	Int	Pk, not null
employee_id	Int	Fk
status	Varchar(10)	Not null , CHECK (status IN ('Draft','Processed','Paid'))
payrollPeriod	Date	Not null
grossSalary	Decimal(12,2)	Not null
totalDeduction	Decimal(12,2)	Not null
netSalary	Decimal(12,2)	Not null
paymentDate	Date	Not null

9. Payroll Details

Column	Type	Constraint
payrollDetail_id	Int	Pk , unique
Payroll_id	Int	Fk , not null
componentName	Varchar(100)	Not null
componentType	Varchar(100)	Not null, CHECK (componentType IN ('Earning','Deduction'))
amount	Decimal(12,2)	Not null

Employee Management & Payroll System

Table Relationships

Parent	Child	FK Column	Cardinality
Department	Employee	department_id	1 : N
Designation	Employee	designation_id	1 : N
Role	Employee	role_id	1 : N
Employee	Employee	manager_id	1 : N (self-referencing)
Employee	Attendance	employee_id	1 : N
Employee	Leave	employee_id	1 : N
Employee	Leave	approvedBy → employee_id	1 : N
Employee	SalaryStructure	employee_id	1 : N
Employee	Payroll	employee_id	1 : N
Payroll	PayrollDetails	payroll_id	1 : N

Payroll Calculation Mechanism

Basic Salary = SalaryStructure.basicSalary (active row where $\text{effectiveFrom} \leq \text{payrollPeriod} \leq \text{effectiveTo}$ / current)

HRA = SalaryStructure.hra

Allowances = SalaryStructure.allowances

HRA and Allowances are paid in full regardless of attendance — only Basic Salary is attendance-linked.

Attendance Adjustment:

Employee Management & Payroll System

workingDaysInMonth = total working days in payrollPeriod (fixed assumption, e.g. 26)

unpaidDays = count of Attendance.status = 'Absent' + Leave rows where *leaveType* = 'Unpaid' and status = 'Approved'

perDayBasic = BasicSalary / workingDaysInMonth

unpaidLeaveDeduction = perDayBasic × unpaidDays

Totals:

Total Deductions = SalaryStructure.deductions + unpaidLeaveDeduction

Gross Salary = BasicSalary + HRA + Allowances

Net Salary = Gross Salary – Total Deductions

PayrollDetails rows = one row per component (BasicSalary, HRA, Allowances, UnpaidLeaveDeduction, and any itemized SalaryStructure.deductions)

componentType = 'Earning' | 'Deduction'